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Roll No

IT-4005-CBGS

B.E. IV Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Computer Architecture

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Draw Von Neumann model of computer and explain all the subsystems of computer.
b) What do you mean by SHIFT micro operation? Explain all three types of SHIFT micro-operation.
2. a) Draw one stage of arithmetic unit with its function table.
b) Explain all the CPU registers with their size and functions. Explain how instructions are executed in a computer?
3. a) Explain signed-magnitude, signed 1's and 2's complement representations of negative numbers.
b) Explain the process of multiplication using BOOTH method. Solve -5×2 using Booth method.
4. a) Differentiate hardwired and microprogrammed control units. Define microprogram sequencer.
b) What do you mean by zero, one two and three address instructions? Give suitable examples.

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5.
 - a) What is Cache? Explain the principle of “Locality of references”. Enlist and explain page replacement algorithms.
 - b) Explain hit ratio. Generate expression for hit ratio. Drive an expression for “effective access time (t_e)” if t_1 and t_2 are access time of cache and main memory respectively. h_1 is the hit ratio of cache.
6.
 - a) What is Mapping? Name all the types of cache mapping and explain any one in detail.
 - b) What are the advantages of parallel processing? Define the terms: Pipeline, speed up ratio and memory interleaving.
7.
 - a) Write a short note on virtual memory.
 - b) What is multiprocessor? Explain the characteristics of multiprocessors.
8. Write short notes on any two of the followings.
 - a) Addressing modes
 - b) Hardwired control unit
 - c) DMA data transfer
 - d) Daisy chaining
